What this is

A controller that drives the carriage and the cross slide with stepper motors, keeping them in step with the spindle. It does the job of change gears and a lead screw, and adds automatic operations on top: mark the corners of the cut and it will make the passes for you.

Limits	Most jobs need you to mark the corners of the cut first — how far along, and how deep in — with the four limit keys. They are not guard rails: they *are* the cut.
Passes	A cut taken in stages. You say how many; it divides the depth between them and lifts the tool clear on the way back.
Feed	How far the tool moves per turn of the spindle. On a thread that is the pitch; on a turning cut it sets the finish.
It moves itself	Keep clear, know where the stop key is, and try anything new with the tool well away from the work.

Choosing what it does

Each job has a key, shown in the table below. Two keys have a second job behind them: press the same key again to reach it, marked **×2**.

M steps through the rest: fourth axis (if fitted) → ball and dish → G-code → steady feed → slotting → back to the gearbox

Job	Shows	Key	What it is for	Limits	What you can adjust
Gearbox	—		The carriage feeds along by itself as the spindle turns — what replaces change gears.	not needed	–
Cross-feed	XGEAR	×2	The same, driving the cross slide across the face. Powered facing.	not needed	–
Turning	TURN		Reduces a diameter over several passes, lifting the tool clear on each return.	all four	Passes · Spring passes · Clearance
Facing	FACE		Cleans up or shortens the end of the work, in several passes.	all four	Passes · Spring passes · Clearance
Parting	CUT		Cuts the work off, or cuts a groove. Can back out to clear the swarf.	up, down	Passes · Peck depth · Clearance
Threading	THRD		Cuts a screw thread, deeper each pass. Pick it from the list of 52 and the feed is set.	all four	Passes · Spring passes · Flank infeed · Clearance · Thread list
Tapered thread	TPR	×2	A thread on a taper rather than a parallel bar — pipe fittings, NPT and BSPT.	all four	as threading, plus its own taper
Taper	CONE		The tool moves in as it travels along, at a ratio you set. Feed in by hand.	not needed	taper ratio, asked at the start
Ball and dish	ELLI	M	Rounds the end into a ball, or hollows it into a dish.	all four	Passes
Slotting	SLOT	M	The lathe as a shaper: spindle still, tool stroking. Keyways and flats.	all four	Passes · Stroke shortening
Steady feed	ASY	M	A set feed rate with the spindle out of it. Handy for polishing.	not needed	–
G-code	GCODE	M	Runs a program written on a computer and sent over.	not needed	–
Fourth axis	A1	M	Drives a fourth motor, if you have fitted one.	not needed	–

Running one of the automatic jobs

1		Set the feed, then move the tool to each corner of the cut and press the matching limit key. On a thread the **up** limit is full thread depth — see below.
2		The bottom line reads **ON to set up 3 steps**. Press play to begin.
3		It asks a short series of questions. **1/3** tells you which one you are on and how many there are. **<>** means the arrow keys change the answer; otherwise type a number. Play accepts and moves on.
4		The last question is **Go?**, and it shows how far the tool will travel to reach the starting corner before it cuts anything. Play starts the job; stop takes you back to the beginning.
5		While it runs the line reads **Pass 2 of 6**. Play skips to the next pass if you have taken enough off; stop, or any jog key, ends the job.

How deep should a thread go? The controller sets the pitch from its thread list, but it cannot know the depth — that is what the up limit tells it, and getting it wrong is the usual reason a first thread comes out badly. For an ordinary 60° thread the depth is **0.6134 × pitch** measured inwards from the surface, so an M10×1.5 goes 0.92mm deep. If the cross slide is set to read diameters rather than distance from centre, put in twice that.

Reading the screen

```
THRD* off step 1.00
Pitch 1.50mm x2
Z 12.345 X 25.400
3/3 Go Z-5.00mm?
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Top	Which job is selected, whether it is running, a small mark for each limit you have set, and how far one tap of a jog key moves.
Second	The feed, and the number of starts if the thread has more than one.
Third	Where the tool is. A slashed X means that figure is a diameter.
Bottom	Everything else — the setup questions, how many passes are done, why something was refused, and whichever readout you have turned on.

The **display** key changes what the bottom line shows when it is otherwise idle: spindle angle, then spindle speed alongside the cutting speed at the tool, then big position digits you can read from a few feet away, then nothing.

Setting up a new machine

Flash the firmware, then choose millimetres or inches with the measure key. The controller has to be told about your lathe before it can be accurate, and it can measure most of that itself. **Hold** the settings key, choose **Calibration**, press play, and work down this list — it begins with the routines that move nothing and ends with the one that deliberately stalls a motor.

#	Routine	Does it move?	What it is for
1	**Input tester**	nothing moves	Press every key and check it registers. Confirms the panel works before anything is asked to move.
2	**Encoder signal**	nothing moves	Watches the spindle sensor for electrical interference. Should sit at 100 while the spindle turns.
3	**Encoder PPR**	you turn the chuck	Turn the spindle a set number of times by hand; it works out how finely the sensor reads.
4	**Encoder direction**	nothing moves	Turn the spindle forwards. If the number counts down instead of up, this flips it.
5	**Z / X direction**	the axis moves	Checks each axis travels the way you expect, before anything is measured against it.
6	**Z / X screw pitch**	the axis moves	Moves a set distance; you measure what it actually moved, and it corrects itself to match.
7	**Z / X backlash**	the axis moves	Finds the slack in each screw against a dial indicator, so it can be taken up automatically.
8	**Z / X travel limit**	you jog it	Wind each axis to its ends so the controller knows how far it is allowed to go.
9	**Z / X max speed**	the axis moves	Speeds up until the motor stalls, then backs off. Leave this one until last.

Then save what you have done — there is no undo. Connect a computer over USB, send a single **$**, and it prints every setting as a list of lines you can keep in a text file. Paste them back to restore. The same file can be downloaded from the web page.

Settings that describe your lathe

Hold the settings key to open the menu. Everything here describes the machine itself, and you will mostly set it once. Anything to do with one particular job — how many passes, how much clearance — lives on that job's own page instead, which is a **short** press of the same key.

How things behave	
X readout	Whether the cross slide shows how far it is from centre, or the diameter that gives — twice as much. Lathe work is usually quoted as a diameter.
Retract distance	How far the tool jumps clear when you press the retract key.
Manual step time	How long one tap of a jog key takes.
Step rest	A pause between stepped moves.
Spindle divisions	Turns the angle display into an indexer for cutting flats or keyways. 0 is off; 6 gives you a hexagon.
Index tolerance	How close to a mark counts as being on it.
Constant speed	The single switch for the cutting-speed helper below.
Material	What you are cutting. Sets a sensible cutting speed for you.
Tool	High speed steel or carbide. Carbide takes roughly three times the speed.
Manual speed	Your own figure, used when Material is set to Manual.
Spindle max rpm	The fastest your lathe runs, so it never suggests more.

Spindle sensor	
Encoder PPR	How many pulses it gives per turn. Usually printed on the body.
Direction	Flips the counting direction without rewiring anything.
Spindle / sensor pulley	Tooth counts, if the sensor is belt driven rather than straight off the spindle.
Divider	Ignores very small movements, for a sensor that twitches at rest.
Dead-band	How far the spindle must turn back before the tool follows it.
Dead-band shape	One-way suits a worn lead screw; symmetric suits a noisy sensor.
Glitch filter	Ignores electrical spikes. Set too high it starts missing real pulses.

Each axis · Z, X and the fourth	
Invert direction	If the axis travels the wrong way.
Backlash	The slack in the screw, taken up automatically whenever it reverses.
Lead screw pitch	How far the axis moves for one turn of its screw.
Motor steps/rev	From the motor and its driver settings together.
Motor / screw pulley	Tooth counts, if the motor drives through a belt.
Start speed	How gently it sets off before speeding up.
Max speed	The fastest it will travel.
Acceleration	How briskly it gets up to that speed.
Max travel	How far the axis can physically go, used as a safety net.
Hold when idle	Whether the motor stays locked when stopped. Turn off if it runs hot.
Fitted / Rotary	Fourth axis only: whether it exists, and whether it turns or slides.

Handwheels	
Fitted 1 / 2	Whether each handwheel is connected.
Drives 1 / 2	Which axis it winds.
Invert 1 / 2	If it winds the wrong way.
Pulses per rev	Of the handwheel itself.
Min pulse width	Ignores electrical noise on its wiring.
Dead-band	Stops a wobble being read as a change of direction.

Joystick	
Fitted	Uses all six spare terminals, so nothing else can share them.
Debounce	Ignores the brief chatter a switch makes as it closes.

WiFi	
Enabled	Makes the controller its own small network. Off means the radio is not on at all.
Access PIN	The password, exactly 8 digits. Change it from the default.

Every thread it knows

Choose one from the machine and the feed is set for you. **Depth** is how far in from the surface the tool has to go, in millimetres, measured on the radius — that is what the up limit is for, and it is the one thing the controller cannot work out for itself. If the cross slide is set to read diameters, put in twice the figure shown.

Thread	Pitch	Depth
Metric · 60°		
M3 coarse x0.5	0.50mm	0.31
M4 coarse x0.7	0.70mm	0.43
M5 coarse x0.8	0.80mm	0.49
M6 coarse x1.0	1.00mm	0.61
M8 coarse x1.25	1.25mm	0.77
M8 fine x1.0	1.00mm	0.61
M10 coarse x1.5	1.50mm	0.92
M10 fine x1.25	1.25mm	0.77
M12 coarse x1.75	1.75mm	1.07
M12 fine x1.5	1.50mm	0.92
M14 coarse x2.0	2.00mm	1.23
M16 coarse x2.0	2.00mm	1.23
M16 fine x1.5	1.50mm	0.92
M20 coarse x2.5	2.50mm	1.53
M24 coarse x3.0	3.00mm	1.84

Thread	Pitch	Depth
Unified · UNC and UNF · 60°		
1/4-20 UNC	20tpi	0.78
1/4-28 UNF	28tpi	0.56
5/16-18 UNC	18tpi	0.87
5/16-24 UNF	24tpi	0.65
3/8-16 UNC	16tpi	0.97
3/8-24 UNF	24tpi	0.65
7/16-14 UNC	14tpi	1.11
1/2-13 UNC	13tpi	1.20
1/2-20 UNF	20tpi	0.78
5/8-11 UNC	11tpi	1.42
5/8-18 UNF	18tpi	0.87
3/4-10 UNC	10tpi	1.56
3/4-16 UNF	16tpi	0.97
1-8 UNC	8tpi	1.95

Thread	Pitch	Depth
British parallel pipe · 55°		
G1/8 BSPP 28tpi	28tpi	0.58
G1/4 BSPP 19tpi	19tpi	0.86
G3/8 BSPP 19tpi	19tpi	0.86
G1/2 BSPP 14tpi	14tpi	1.16
G3/4 BSPP 14tpi	14tpi	1.16
G1 BSPP 11tpi	11tpi	1.48
Trapezoidal · 30°		
Tr8x1.5 trap	1.50mm	1.00
Tr10x2 trap	2.00mm	1.25
Tr12x3 trap	3.00mm	1.75
Tr16x4 trap	4.00mm	2.25
Tr20x4 trap	4.00mm	2.25
ACME · 29°		
1/4-16 ACME	16tpi	1.05
5/16-14 ACME	14tpi	1.16
3/8-12 ACME	12tpi	1.31
1/2-10 ACME	10tpi	1.52
5/8-8 ACME	8tpi	1.84
3/4-6 ACME	6tpi	2.37
1-5 ACME	5tpi	2.79
American taper pipe · 60° taper		
1/8-27 NPT	27tpi	0.75
1/4-18 NPT	18tpi	1.13
3/8-18 NPT	18tpi	1.13
1/2-14 NPT	14tpi	1.45
3/4-14 NPT	14tpi	1.45

Depths are for a full-form thread on the outside of a bar. An internal thread is cut to the same depth outwards from the bore. Trapezoidal and ACME include the usual root clearance; taper pipe threads are deeper than they look because the form is truncated rather than pointed.

Cutting one of them

1	Pick the thread from the machine's own list — press the settings key briefly while in threading, then choose it. That sets the feed.
2	Touch the tool on the outside of the bar and zero the cross slide. Set the **down** limit there: that is the surface.
3	Wind in by the depth from the table and set the **up** limit. That is full thread depth. In diameter readout, wind in twice the figure.
4	Set the two length limits, leaving room at the far end for the tool to come out — a run-out groove if the thread ends against a shoulder.
5	Six to ten passes suits most threads, plus a spring pass or two. More passes means a lighter cut each time and a better finish.

Cutting speeds

For setting the spindle yourself. Metres per minute measured at the surface of the work, not the spindle speed — the two are only the same on a one-metre bar. Conservative starting points for turning rather than limits: depth of cut, how rigid the setup is and whether you are using coolant all matter as well. Work up from these while the finish and the swarf still look happy, and drop to roughly half for parting off.

Material	HSS	Carbide
Aluminium	70 m/min	200 m/min
Brass	60 m/min	180 m/min
Bronze	40 m/min	120 m/min
Copper	50 m/min	150 m/min
Cast iron	25 m/min	90 m/min
Mild steel	30 m/min	120 m/min

Material	HSS	Carbide
Alloy steel	20 m/min	90 m/min
Tool steel	15 m/min	60 m/min
Stainless	15 m/min	60 m/min
Titanium	10 m/min	40 m/min
Plastic	100 m/min	250 m/min

Spindle speed for that cutting speed

rpm = 318 × cutting speed ÷ diameter in mm. Read it off here instead: the diameter you are cutting down the side, the speed from the table above across the top. Round down to whatever your lathe actually offers.

Ø mm	20	30	60	90	120	200
6	1060	1590	3180	4770	6370	10610
10	640	950	1910	2860	3820	6370
16	400	600	1190	1790	2390	3980
20	320	480	950	1430	1910	3180
25	250	380	760	1150	1530	2550
32	200	300	600	900	1190	1990
40	160	240	480	720	950	1590
50	130	190	380	570	760	1270
63	100	150	300	450	610	1010
80	80	120	240	360	480	800
100	65	95	190	290	380	640

Top row is metres per minute. Anything above your lathe's top speed simply means run it flat out — small diameters usually want more than the machine has.

Taper ratios

What to type when the taper or tapered-thread setup asks for a ratio. It is the change in **diameter** per unit of length — (large – small) ÷ length — so a taper quoted as 1 in 16 is entered as 0.0625.

Taper	Enter	Included	Where
Morse 1	0.0499	2° 51'	tailstock and spindle tooling
Morse 2	0.0500	2° 52'	the commonest on small lathes
Morse 3	0.0502	2° 53'	–
Morse 4	0.0519	2° 59'	–

Taper	Enter	Included	Where
Morse 5	0.0526	3° 1'	–
NPT and BSPT	0.0625	3° 35'	taper pipe threads, 1 in 16
1 in 10	0.1000	5° 43'	–
1 in 20	0.0500	2° 52'	–

Working from an angle instead: ratio = 2 × tan(half the included angle). Cutting a taper on the outside and a matching socket needs the same ratio for both — the mode does not care which way round you are cutting.

Using it from a phone

1	Settings → WiFi → **Enabled** = yes. Change the **Access PIN** from the default first — anyone who can join the network can change the machine's settings.
2	Join the **NanoEls-H4** network with that PIN, then open **192.168.4.1**. The panel shows the address.
3	Every setting is there, quicker to type than on the panel, with position and speed live. It also shows how clean the spindle sensor's signal is — leave it open through a job and check afterwards.
4	New firmware installs from the same page, with no computer at the machine. Only while it is stopped.